

Objectives

- Explain React components
- Identify the differences between components and JavaScript functions
- Identify the types of components
- Explain class component
- Explain function component
- Define component constructor
- Define render() function

In this hands-on lab, you will learn how to:

- Create a class component
- Create multiple components
- Render a component

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **30 minutes**.

Create a react app for Student Management Portal named StudentApp and create a component named Home which will display the Message “Welcome to the Home page of Student Management Portal”. Create another component named About and display the Message “Welcome to the About page of the Student Management Portal”. Create a third component named Contact and display the Message “Welcome to the Contact page of the Student Management Portal”. Call all the three components.

1. Create a React project named “StudentApp” type the following command in terminal of Visual studio:

```
C:>npx create-react-app StudentApp
```

2. Create a new folder under Src folder with the name “Components”. Add a new file named “Home.js”

3. Type the following code in Home.js

```
import React, {Component} from 'react';

import class Home extends Component{
  render(){
    <div>
    <h3> Welcome to the Home Page of Student Management Portal </h3>
    </div>
  }
}
```

4. Under Src folder add another file named "About.js"
5. Repeat the same steps for Creating "About" and "Contact" component by adding a new file as "About.js", "Contact.js" under "Src" folder and edit the code as mentioned for "Home" Component.
6. Edit the App.js to invoke the Home, About and Contact component as follows:

```
WEEK-4 > 02.Reactjs-hol > studentapp > src > App.js > default
1  import Home from './components/Home';
2  import About from './components/About';
3  import Contact from './components/Contact';
4
5  function App() {
6    return (
7      <div>
8        <Home />
9        <About />
10       <Contact />
11     </div>
12   );
13 }
14
15 export default App;
```

7. In command Prompt, navigate into StudentApp and execute the code by typing the following command:

```
C:\studentapp>npm start
```

8. Open browser and type "localhost:3000" in the address bar:



Welcome to the Home page of Student Management Portal

Welcome to the About page of Student Management Portal

Welcome to the Contact page of Student Management Portal

A React component is a reusable piece of code that represents a part of the user interface (UI). Components help divide the UI into independent and reusable parts.

Example: Header, Footer, Navigation Bar, Login Form.

There are two types of React components:

- Class Components
- Function Components

A Class Component is a JavaScript class that extends `React.Component`. It contains a `render()` method that returns JSX.

A constructor is a special method in a class component that is executed when the component is created. It is mainly used to initialize the component's state and bind methods.

A Function Component is a JavaScript function that returns JSX. It is simpler than a class component.

The `render()` method is a mandatory method in a class component. It returns the JSX that should be displayed on the webpage.